

AI ADOPTION & WORKFORCE ANALYTICS DASHBOARD

Data Analytics Project Documentation

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Excel | SQL | Power BI | Python | Github

OBJECTIVE

This objective of this project is to analyze workforce trends in the Artificial Intelligence domain using data analytics techniques. The project aims to identify salary patterns, industry-wise AI adoption, required skills, automation risks, remote work opportunities, and growth projections. Interactive dashboards were developed in Power BI to support business decision-making.

PROBLEM STATEMENT

Artificial intelligence is transforming industries across the world. Organizations require meaningful insights to understand workforce trends , salary distribution, skill demand, and future job growth. This project analyzes an AI workforce dataset to convert raw data into actionable business insights through SQL, Python and Power BI.

DATA DESCRIPTION

ATTRIBUTE	DESCRIPTION	
Job Title	Employee designation	
Industry	Industry sector	
Company Size	Small, Medium, Large	
Location	Job location	
Salary USD	Annual Salary	
AI Adoption Level	Low, Medium, High	
Required Skills	Skills required	
Remote Friendly	Yes / No	
Growth Projection	Stable / Growth / Decline	
Automation Risk	Low / Medium / high	

DataSet Size

Record: 500

Columns: 10

TOOLS & TECHNOLOGIES

Excel

SQL

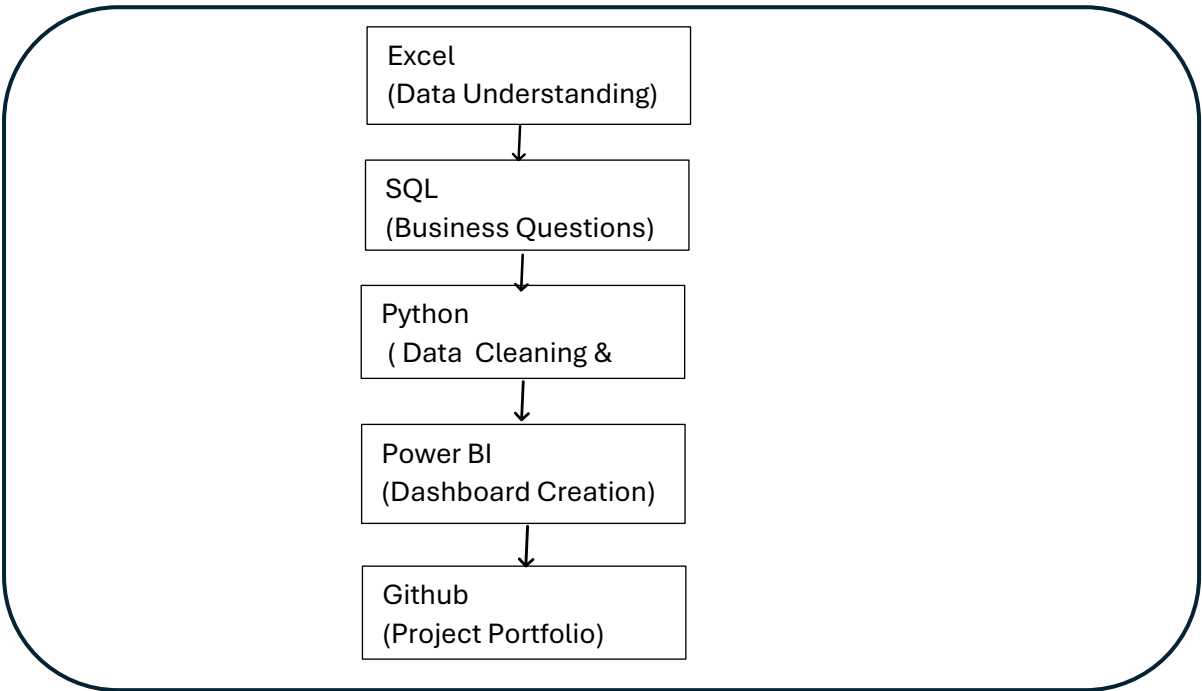
Python

Power

Github

Tools	Purpose
Excel	Initial data understanding
SQL	Business Questions
Python	Data Cleaning & EDA
Pandas	Data Manipulation
NumPy	Numerical analysis
Matplotlib	Data Visualization
VsCode	Python development
Power BI	Dashboard creation
Power Query	Data transformation
Github	Project hosting

PROJECT WORKFLOW



SQL BUSINESS QUESTIONS

PURPOSE

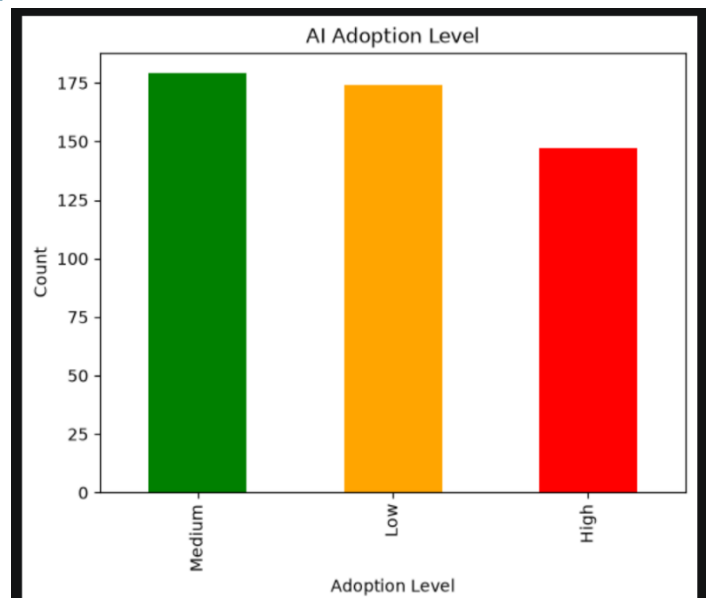
SQL used to answer business questions and explore patterns within the dataset before visualization.

Business Questions	Purpose
Total number of jobs	Dataset Overview
Average salary by industry	Salary comparison
Jobs by company size	Workforce distribution
AI adoption level	Adoption analysis
Remote - Friendly	Work mode analysis
Growth Projection	Future Trends

Exploratory Data Analysis (EDA)

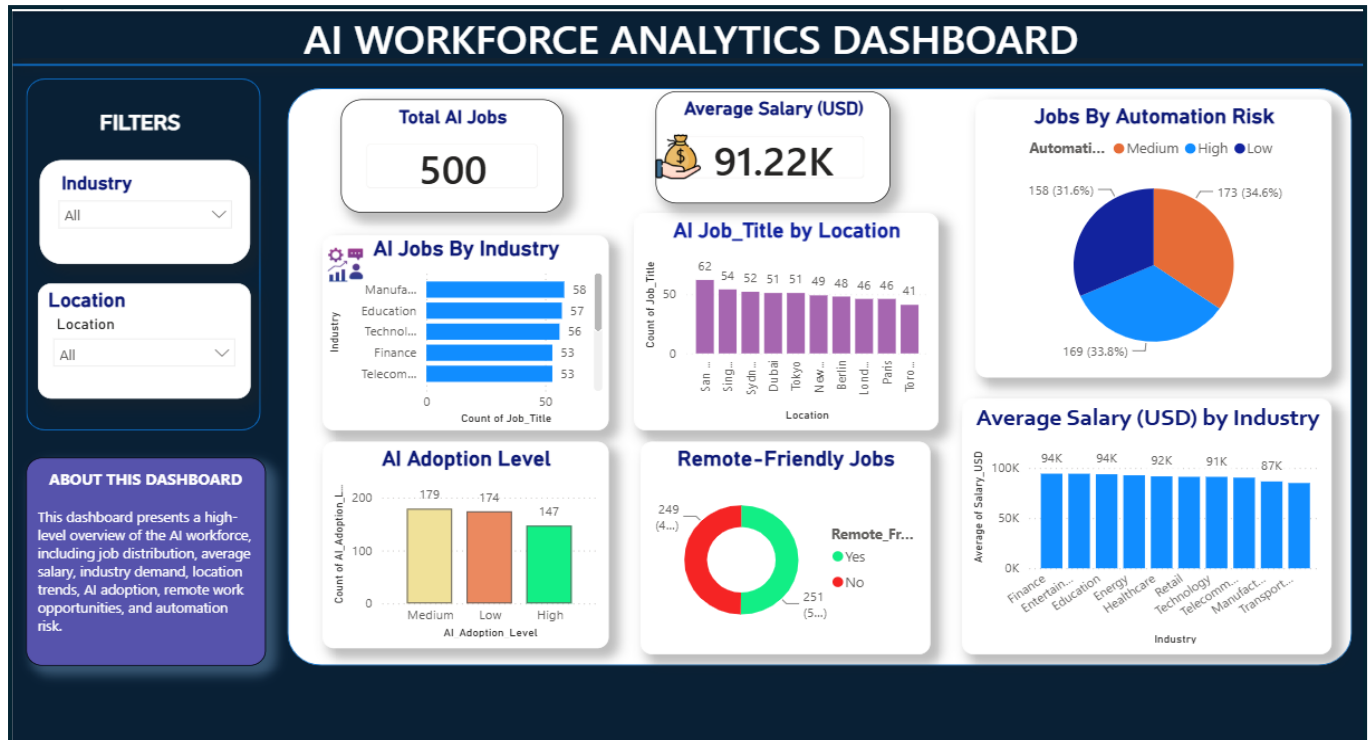
Tasks Performed

- Imported dataset
- Checked data types
- Identified missing values
- Checked duplicate records
- Generated summary statistics
- Created Visualizations



Python help validate the dataset and understand its structure before creating Power BI dashboards.

DASHBOARD 1 OVER VIEW



PURPOSE

This dashboard provides an overall summary of AI workforce dataset. It presents key metrics and high-level trends to help users quickly understand the dataset.

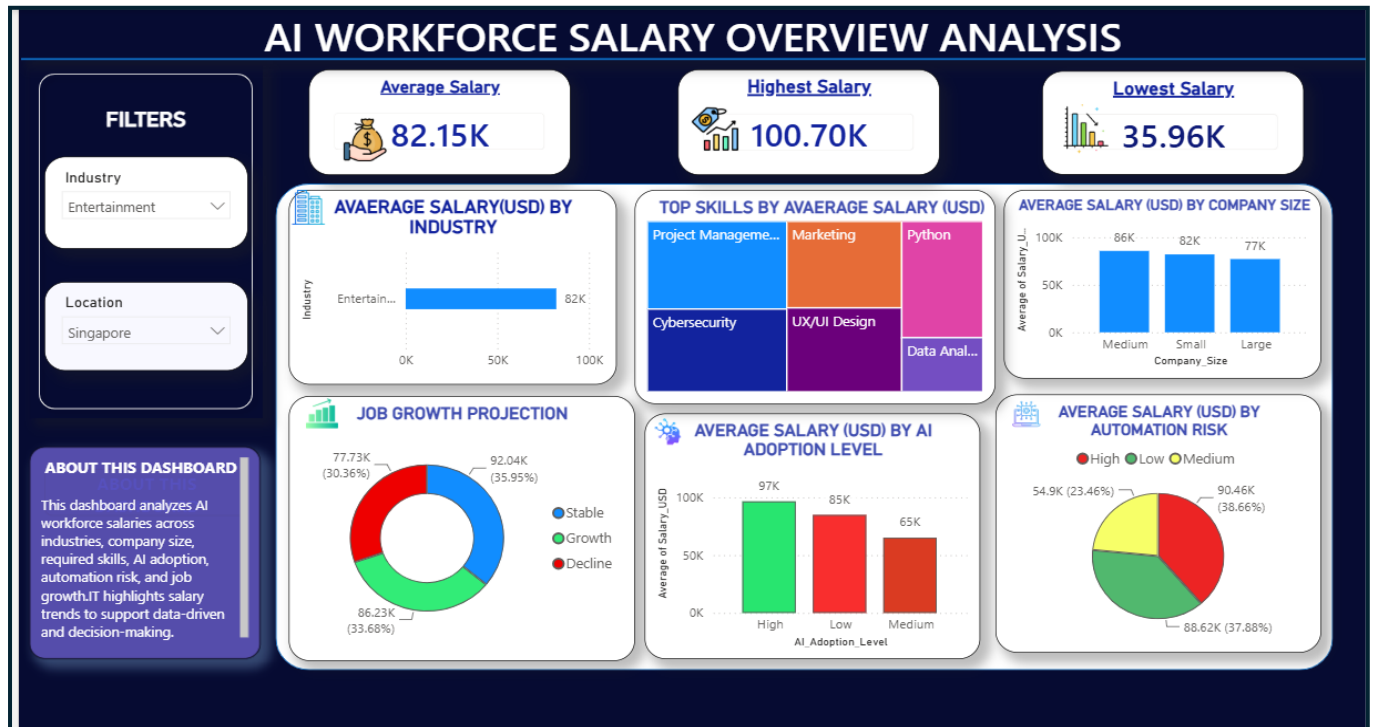
KEY VISUALS

- KPI Cards
- Clustered Bar Chart
- Clustered Column Chart
- Pie Chart
- Clustered Column Chart
- Donut Chart
- Clustered Column Chart

BUSINESS VALUE

- Provides a quick overview of workforce trends and supports high-level business decision – making.

DASHBOARD 2 SALARY ANALYSIS



PURPOSE

This dashboard analyzes salary trends across different industries, company sizes, skills, and AI adoption levels.

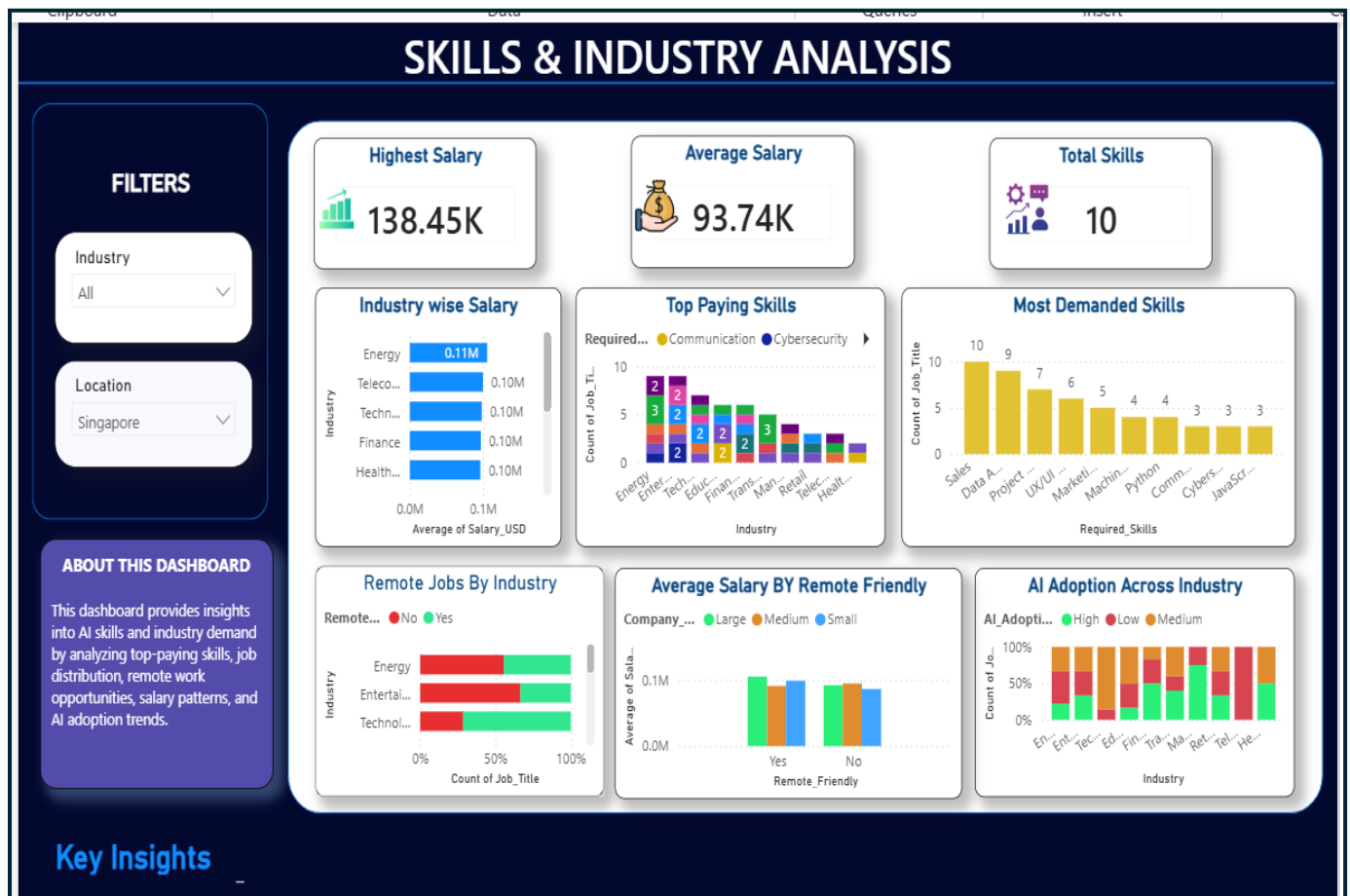
KEY VISUALS

- KPI Cards
- Clustered Bar Chart
- Treemap
- Clustered Column Chart
- Donut Chart
- Clustered Column Chart
- Pie Chart

BUSINESS VALUE

Helps understand the factors influencing salaries and supports compensation benchmarking.

DASHBOARD 3 SKILLS & INDUSTRY ANALYSIS



PURPOSE

This dashboard explores the relationship between required skills and different industries.

KEY VISUALS

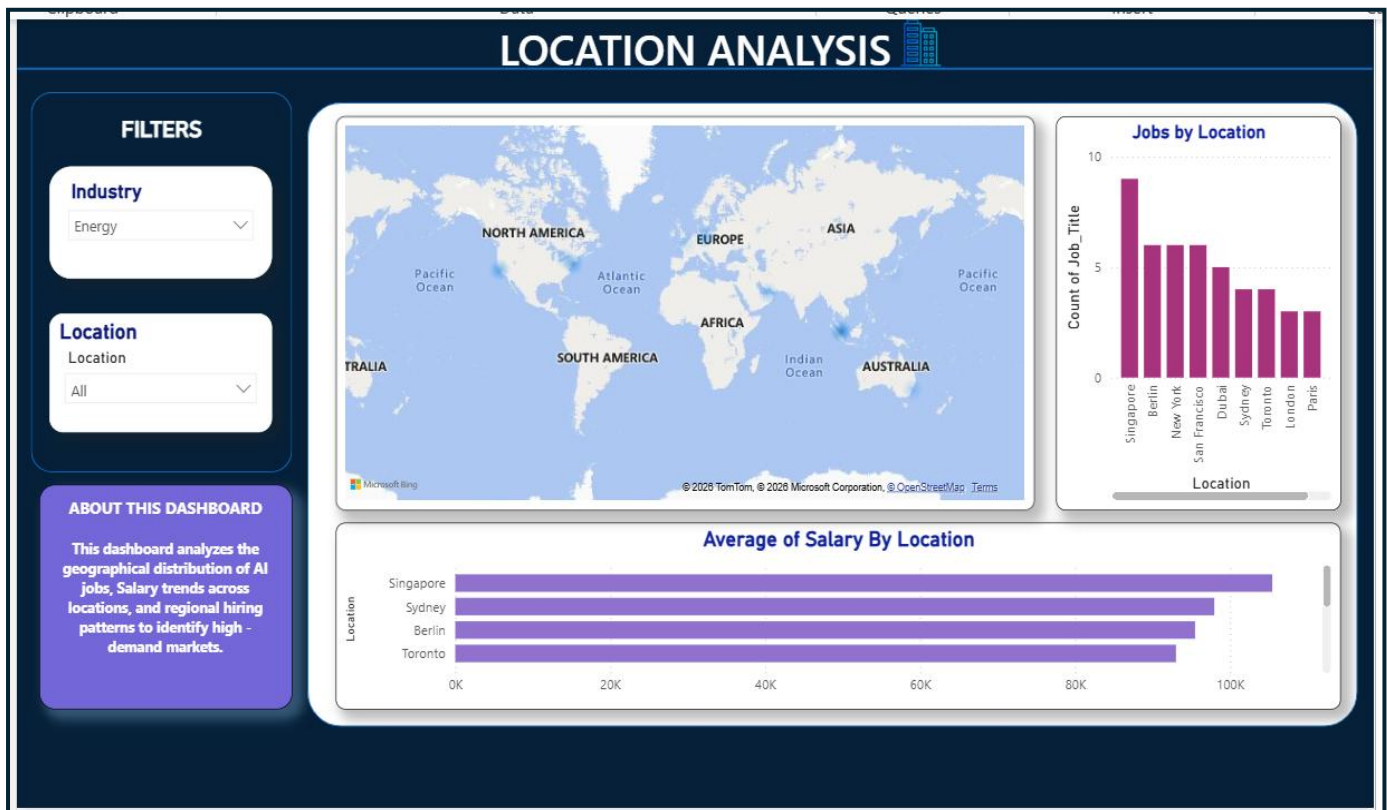
- KPI Cards
- Clustered Bar Chart
- Stacked Column Chart
- Clustered Column Chart

- 100% Stacked Bar Chart
- Clustered Column Chart
- 100% Stacked Column Chart

BUSINESS VALUE

Helps identify in-demand skills and understand industry-specific workforce requirements.

DASHBOARD 4 LOCATION ANALYSIS



PURPOSE

This distribution analyzes geographical distribution of AI jobs and salary variations across locations.

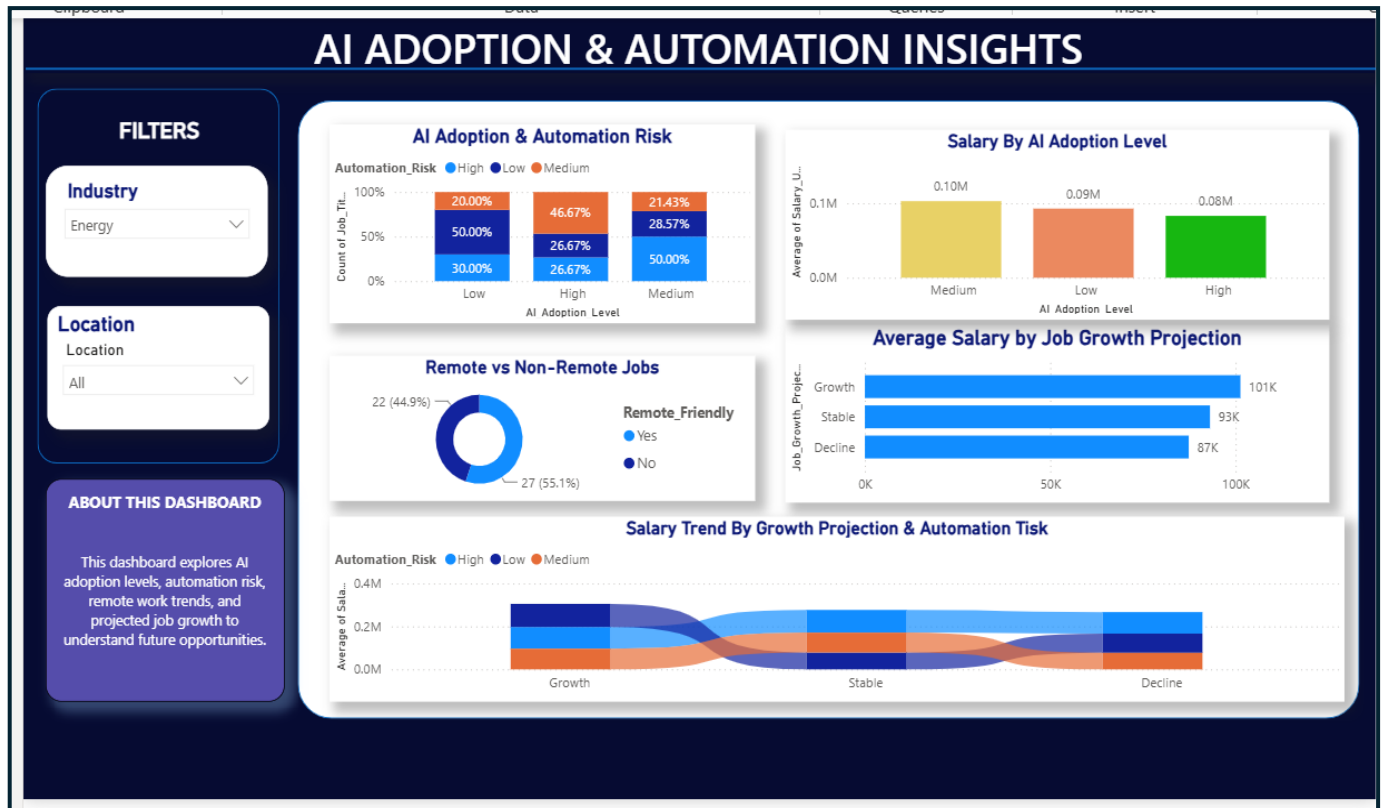
KEY VISUALS

- Filled Map
- Clustered Column Chart
- Stacked Column Chart

BUSINESS VALUE

Helps compare workforce trends across different locations and supports location-based analysis.

DASHBOARD 5 AI ADOPTION & FUTURE WORKFORCE TRENDS



PURPOSE

- This dashboard focuses on AI adoption, automation risk, and future workforce growth.

KEY VISUALS

- 100% Stacked Column Chart
- Clustered Bar Chart
- Donut Chart
- Clustered Bar Chart
- Ribbon chart

BUSINESS VALUE

Supports workforce planning by analyzing AI adoption, automation risk, salary trends, remote work opportunities, and future employment growth.

KEY INSIGHTS

- The average salary across the AI workforce dataset is approximately USD **91,222**.
- The **Finance** industry records the highest average salary among the industries analyzed.
- Organizations with **Low AI Adoption** have the highest average salary in this dataset, indicating salary depends on multiple factors.
- Remote-friendly and non-remote jobs are almost equally distributed, showing a balanced work model.
- Job growth projections are nearly evenly distributed across **Growth, Stable and Decline** Categories.
- **New York** offers the highest average salary among the locations included in the dataset.

CONCLUSIONS

This project successfully analyzed AI workforce trends using Excel, SQL, Python, and Power BI. Through data exploration, SQL-based business analysis, and interactive dashboards, the project identified valuable insights related to salary distribution, industry trends, AI adoption, automation risk, remote work opportunities, and job growth. The findings demonstrate how data analytics can transform raw data into meaningful business insights that support informed decision-making. This project also strengthened practical skills in data analysis, visualization, and business intelligence while showcasing an end-to-end analytics workflow.

DATA SOURCE

DataSet: AI Job Market Insights Dataset

Source: Kaggle

The dataset contains 500 AI jobs records covering industries, job titles, salaries, locations, AI adoption levels, automation risk, required skills, company size, remote work, and job growth projections. It was used solely for educational and analytical purposes in this project.